

2 (short)

A long-span timber roof structure is constructed as a three-pinned arch as shown in Fig. 2. The arch has a parabolic geometry defined by:

$$y = d \frac{x^2}{L^2}$$

A uniformly distributed load w per unit horizontal length acts over one half of the structure. The self-weight of the structure is negligible. Determine the magnitude of the bending moment at $x = \frac{L}{2}$. [10]

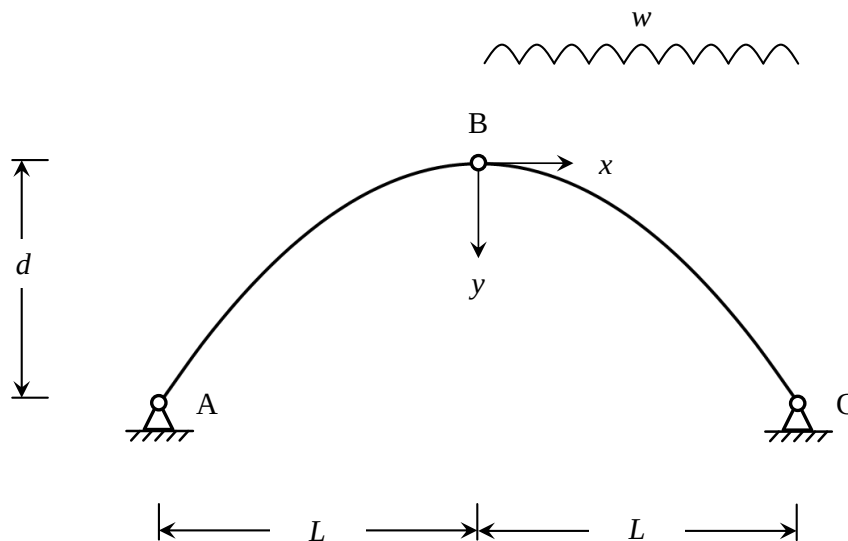


Fig. 2